Introduction to Statistics

Statistics is the branch of mathematics that deals with the collection, organization, analysis, interpretation, and presentation of data. It helps us understand and describe phenomena in the world around us by turning raw data into meaningful information.

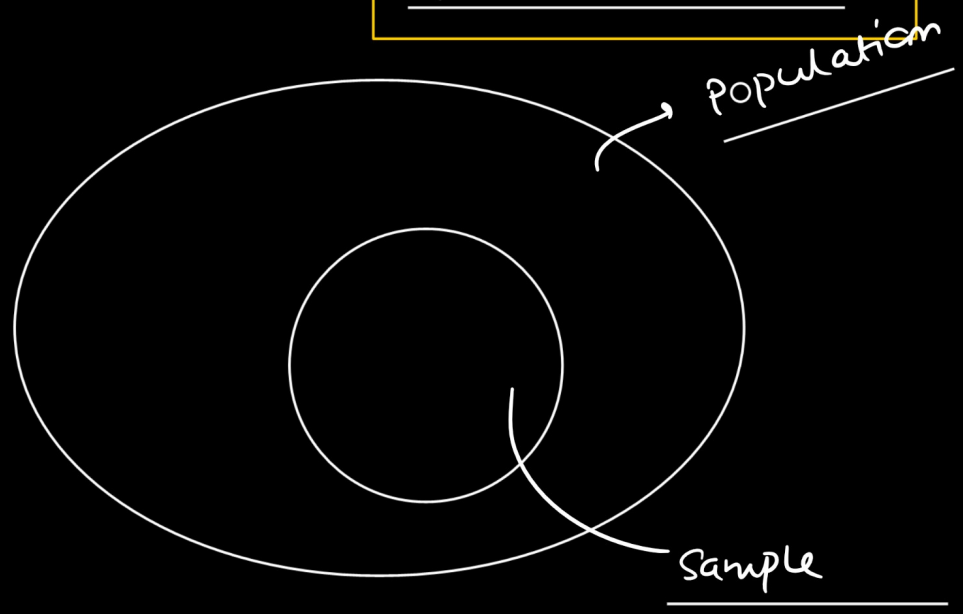
→ Data rules

The Need for Statistics:

Describing Large Populations: We often cannot gather data on every individual in a large population. Statistics allow us to summarize and make sense of data collected from samples, providing insights about the whole population.

Summarizing Data with Samples: Instead of trying to understand everyone or everything, we collect samples. By analyzing these samples, we can infer behaviors, trends, and patterns that are representative of the larger population.

Population & Sample



1. Descriptive Stats → Describe the entire data (Measures of dispersion, central tendency, CLT, Data distribution) → population data

2. Inferential Stats → sample data & statistical tests

(z-test, t-test, ANOVA test,

chi-square test)